UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

a) Why is the machine in zero computer generation known as a mechanical computer?

b) Complete the following table with the appropriate answer.

Mechanical Calculatin Machine Difference Machine Inventor Function

QUESTION A2 **(5 marks)**

Given the following Figure 2.

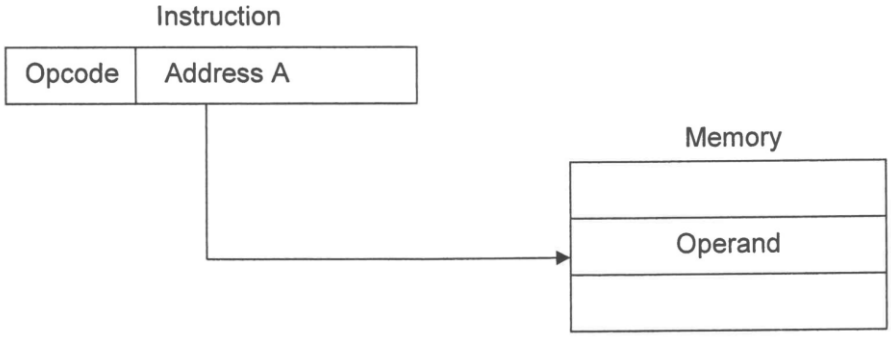


Figure 2

a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

QUESTION A3 **(5 marks)**

a) List THREE (3) characteristics of Symmetric Multiprocessor (SMP).

b) Describe ONE (1) advantage of Symmetric Multiprocessor (SMP) interms of availability.

QUESTION A4 **(5 marks)**

Convert 378510 to binary. Show your complete workflow.

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QUESTION A5 **(10 marks)**

Answer both parts for the same expression. Assume an 8-bit byte.

Expression: -60 - (-12)

a) Calculate -60 - (-12) using sign and magnitude.

b) Calculate -60 - (-12) using 1's and 2's complement.

QUESTION A6 **(10 marks)**

a) Discuss TWO (2) functions of an input/output module.

b) External input/output devices can be broadly categorized into three categories. List TWO (2) of the categories and provide an appropriate example of devices for each categories.

c) Name ONE (1) input/output technique that requires active CPU intervention and ONE (1) technique that does not require CPU intervention.

QUESTION A7 **(10 marks)**

System interconnection is a mechanism that provides communication among computer components.

a) What is the size of a bus if the word size is 64 bit?

b) List the TWO (2) types of bus.

c) Illustrate the location of a bus between CPU and memory.

d) Differentiate between system interconnection and CPU interconnection.

e) Discuss ONE (1) drawback each for the different types of bus timing.

QUESTION A8 **(10 marks)**

a) List the phases in the machine cycle,

b) Discuss the function of the different CPU components in the machine cycle.

c) What will happen in the machine cycle if access to the main memory is blocked?

QUESTION A9 **(10 marks)**

a) Describe THREE (3) types of interfaces to external devices.

b) Explain TWO (2) types of interrupt.

QUESTION A10 **(10 marks)**

Question I is based on Figure 3. Computer structure A Main cache L2 Cache L3 CPU cache L1 memory

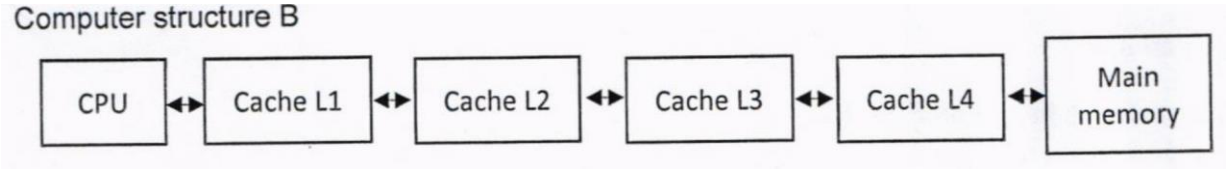


Figure 3: Different computer architectures

a) Explain which computer structure can store the data faster compared to the other. Provide relevant justification to support your answer.

b) Identify the fastest cache level and give an appropriate reason.

c) Vvhat is the main benefit for the cache to use SRAM technology?

d) Discuss TWO (2) methods to update cache and main memory contents to ensure data consistency.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		